19 SOLIDS, LIQUIDS AND GASES

All matter is made up of particles that are continuously moving. The arrangement and movement of the particles determine the properties of the material. In gases, scientists have discovered laws that describe the relationship between pressure, temperature and volume. In this chapter, you will learn what these laws are and how this relationship can be explained in terms of the behaviour of the particles.



▶ Figure 19.1 The botanist Robert Brown (1773–1858) made an observation while studying pollen grains under a microscope that has become known as Brownian motion in his honour. The observation led to the kinetic theory of matter that we use to explain the behaviour of solids, liquids and gases.

LEARNING OBJECTIVES

PHYSICS ONLY

- Explain why heating a system will change the energy stored within the system and raise its temperature or produce changes of state
- Describe the changes that occur when a solid melts to form a liquid, and when a liquid evaporates or boils to form a gas
- Describe the arrangement and motion of particles in solids, liquids and gases
- Practical: obtain a temperature–time graph to show the constant temperature during a change of state
- Know that specific heat capacity is the energy required to change the temperature of an object by one degree Celsius per kilogram of mass (J/kg °C)
- Use the equation:

  change in thermal energy = mass × specific heat capacity × change in temperature

  $$\Delta Q = m \times c \times \Delta T$$
- Practical: investigate the specific heat capacity of materials including water and some solids
- Explain how molecules in a gas have random motion and that they exert a force and hence a pressure on the walls of a container
- Understand why there is an absolute zero of temperature which is –273 °C
- Describe the Kelvin scale of temperature and be able to convert between the Kelvin and Celsius scales
- Understand why an increase in temperature results in an increase in the average speed of gas molecules
- Know that the Kelvin temperature of a gas is proportional to the average kinetic energy of its molecules
- Explain, for a fixed amount of gas, the qualitative relationship between:
  - pressure and volume at constant temperature
  - pressure and Kelvin temperature at constant volume
- Use the relationship between the pressure and Kelvin temperature of a fixed mass of gas at constant volume:

  $$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$
- Use the relationship between the pressure and volume of a fixed mass of gas at constant temperature:

  $$p_1V_1 = p_2V_2$$

SOLIDS, LIQUIDS AND GASES

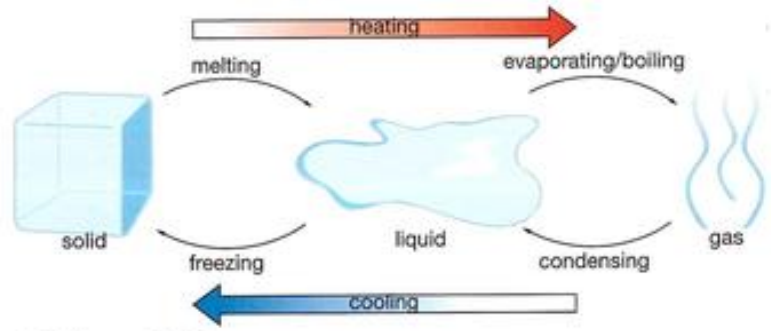
PHYSICS ONLY

We think that all matter is made up of tiny particles that are moving. The way that the particles are arranged and the way that they move determine the properties of a material, such as its state at room temperature or its density.

THE STATES OF MATTER

Some of the properties of a substance depend on the chemicals it is made from. However, substances can exist in different states. The main states of matter are solid, liquid and gas. We are used to finding some substances in each state in everyday life – for example, water is familiar as ice, as water and as steam. There are other substances that we usually see in only one state – for example, we rarely see iron in any state other than solid, or experience oxygen in any state other than as gas.

Substances can change state by the processes of melting, evaporation, boiling, freezing and condensing.



▲ Figure 19.2 Changes of state

PROPERTIES OF THE DIFFERENT STATES OF MATTER

Most of this chapter will be about the properties of gases, but we shall mention solids and liquids for completeness.

SOLIDS

Solids have a definite rigid shape and they are often very dense. The density of a substance is a measure of how tightly packed the particles are.

Some solids have high densities because the particles that they are made from are very closely packed in a regular arrangement. There are strong forces between the particles, which give solid objects their definite shape and, in some materials, a great deal of strength.

Although the particles are held together by strong forces, they can still move. They vibrate about their fixed positions in the solid. When we supply energy to a solid by heating it, the particles vibrate more – they move more quickly. We notice the increase in the kinetic energy of the particles in a substance as an increase in the temperature of the substance.

LIQUIDS

Liquids share a property with gases – they have no definite shape. However, the particles that make up liquids tend to stick together, unlike gas particles. Liquids will occupy the lowest part of any container but gases will expand to fill any container that they are in. Liquids have much greater densities than gases. This is because the particles in liquids are still very close together, like they are in solids. Because the particles in liquids are close together, they still

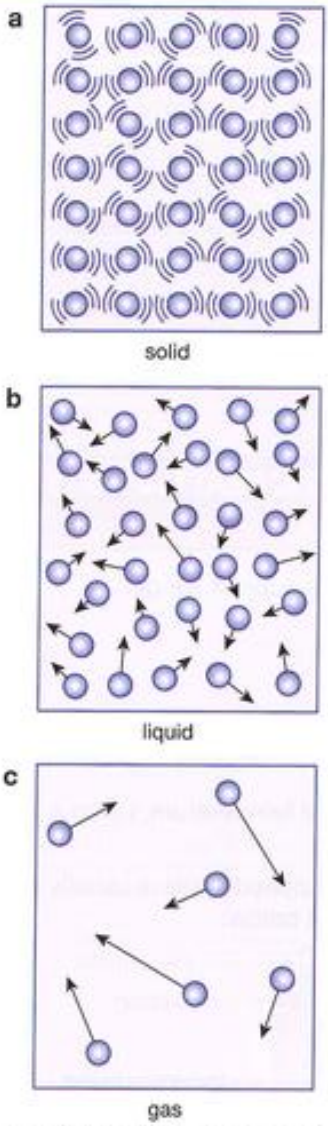
attract one another and hold together. In liquids, there is no fixed pattern and the particles can move around more freely than in solids. As we heat liquids, the movement of the particles becomes more energetic.

GASES

In gases the particles are very spread out, with large spaces between them. This means that the forces holding them together are small. Gases have very low densities and no definite shape. Gases can also be squashed into a smaller space (compressed). Particles of a gas are moving randomly all the time. The particles will bump into anything in the gas, or into the walls of the container, and the forces caused by these collisions are responsible for the pressure that gases exert.

Solids and liquids are very difficult to compress because the particles in them are almost as close together as they can be.

SUMMARY OF THE PROPERTIES OF SOLIDS, LIQUIDS AND GASES



▲ Figure 19.3 The different arrangements of the particles in a solids; b liquids and c gases (not drawn to scale)

Property	Solids	Liquids	Gases
definite shape	yes	no	no
can be easily compressed	no	no	yes
relative density	high	high	low
can flow (fluid)	no	yes	yes
expands to fill all available space	no	no	yes

The particles (molecules) in a solid (Figure 19.3a):

- are tightly packed
- are held in a fixed pattern or crystal structure by strong forces between them
- vibrate around their fixed positions in the structure.

The particles (molecules) in a liquid (Figure 19.3b):

- are tightly packed (still very close together, like in solids)
- are not held in fixed positions but are still bound together by strong forces between them
- move at random with no fixed positions.

The particles (molecules) in a gas (Figure 19.3c):

- are very spread out
- have no fixed positions and the forces between them are very weak
- move with a rapid, random motion.

MEASURING HEAT ENERGY

We have seen that when heat energy is supplied to solids, liquids and gases it increases the kinetic energy of their molecules, which we detect as an increase in temperature. Different substances need different amounts of heat energy to cause the same increase in temperature.

We define the specific heat capacity (s.h.c.), c, of a substance as the amount of energy required to increase the temperature of 1 kilogram of that substance by 1 °C. The unit of s.h.c. is J/kg °C.

SOLIDS, LIQUIDS AND GASES

KEY POINT

Δ, the Greek letter delta, means a change in the quantity that follows it. θ, the Greek letter theta, is used to represent temperature so $\Delta\theta$ (delta theta) means a change in temperature. If we cool something down then $\Delta\theta$ is negative and so, therefore, is ΔQ. This simply means that, to cool something down, we need to remove thermal energy from it.

We use the following equation to work out how much energy is needed to change the temperature of an object by a given amount:

change in thermal energy, ΔQ (joules) = mass, m (kilograms) × specific heat capacity, c × change in temperature, $\Delta\theta$ (°C)

$$\Delta Q = m \times c \times \Delta\theta$$

EXAMPLE 1

If you fill a kettle with 300 g of water at an initial temperature of 15 °C, how much energy is needed to make the water heat up to boiling? The s.h.c. of water is 4200 J/kg °C.

Write down what you know:

m = 0.3 kg (you must work in consistent units)

c = 4200 J/kg °C

and, as the boiling point of water is 100 °C, $\Delta\theta$ = 85 °C

Write down the equation you are using (no marks for this as it is a given equation):

$\Delta Q = m \times c \times \Delta\theta$

Substitute the correct values into the equation (this will normally attract a method mark).

ΔQ = 0.3 kg × 4200 J/kg °C × 85 °C

Now complete the calculation and include the correct unit in your answer.

ΔQ = 107 100 J

EXTENSION WORK

This figure assumes that all the thermal energy provided goes to raise the temperature of the water. In practice some energy will be used to heat up the kettle and some will be lost to the surroundings through thermal radiation, thermal conduction and convection and some water will evaporate before the boiling point is reached.

ACTIVITY 1

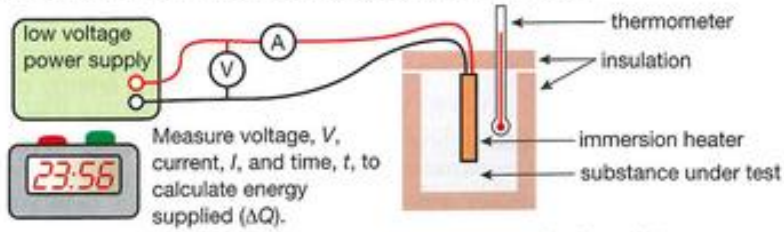
▼ PRACTICAL: INVESTIGATE THE SPECIFIC HEAT CAPACITY OF A SUBSTANCE

Rearranging the equation above to make c the subject we get:

$$c = \frac{\Delta Q}{m \times \Delta\theta}$$

So we must:

- measure the mass, m, in kg of the substance under test using electronic scales
- measure the initial temperature and the final temperature, using a thermometer to find $\Delta\theta$ °C
- determine the amount of thermal energy supplied – this is usually done with an electric immersion heater as shown below.



▲ Figure 19.4 Apparatus for measuring the specific heat capacity of a substance

!

The immersion heater (and, eventually, the metal block) will get hot enough to burn the skin. If used in water, the immersion heater must not have its top immersed.

Figure 19.4 shows typical apparatus for finding the s.h.c. of a metal. A cylinder of the metal is drilled to allow an electric heater and a thermometer to be inserted as shown. Use a balance to find the mass of the cylinder and a thermometer to measure its temperature. Turn the heater on and measure the voltage, V (volts), supplied to the heater, the current drawn, I (amps), and the length of time, t (seconds), the heater is on. Make a note of the highest temperature reached and calculate $\Delta\theta$, the rise in temperature.

The thermal energy supplied by the heater is $\Delta Q = V \times I \times t$ (page 65).

Substitute the measured values for m, ΔQ and $\Delta\theta$ into the equation shown above and calculate the s.h.c.

The insulation is used to cut down heat losses to the surroundings.

This method can be used to measure the s.h.c. of water, with a weighed amount (m) of water in beaker (with a lid and insulation). The water can be heated with an immersion heater and heat energy given to the water calculated in the same way ($\Delta Q = VIt$). The temperature increase is measured with a thermometer.

EXAMPLE 2

In an experiment to measure the specific heat capacity of aluminium, a cylinder of mass 0.5 kg, initially at 20 °C, was supplied with 16 500 J of thermal energy. At the end of this heating the temperature had risen to 50 °C. Calculate the s.h.c. of aluminium.

Write down what you know:

$m = 0.5$ kg

$\Delta Q = 16\,500$ J

temperature rise from 20 °C to 50 °C is $\Delta\theta = 30$ °C

Substitute these values into the given equation:

$$c = \frac{\Delta Q}{m \times \Delta\theta}$$

$$= \frac{16\,500 \text{ J}}{0.5 \text{ kg} \times 30 \text{ °C}}$$

and calculate the value:

$c = 1100$ J/kg °C

As heat will still be lost and some of the heat energy supplied heats the immersion heater and the thermometer, then this answer is larger than the expected value of 900 J/kg °C

EXAMPLE 3

To determine the specific heat capacity of water, 1 litre of water (mass, m, = 1 kg) at a starting temperature of 20 °C is poured into an electric kettle with a power of 2.4 kW. The water in the kettle just reaches boiling point after 2½ minutes. Use this information to calculate the s.h.c. of water.

Do you think your answer will be too big or too small? Give a reason for your answer.

SOLIDS, LIQUIDS AND GASES

As power is the rate of supply of energy you can calculate the energy transferred to the water (ΔQ) using:

energy transferred, ΔQ (J) = power, p (W) × time, t (seconds) (see page 65)

$$\Delta Q = 2400 \text{ W} \times 150 \text{ s}$$
$$= 360\,000 \text{ J}$$

The temperature increase is 80 °C and, conveniently, the mass of water, m, is 1 kg.

Substitute these values into the given equation:

$$c = \frac{\Delta Q}{m \times \Delta\theta}$$

$$= \frac{360\,000 \text{ J}}{1 \text{ kg} \times 80 \text{ °C}}$$

and calculate the value:

$$c = 4500 \text{ J/kg °C}$$

The answer will be higher than the accepted value because of heat losses to the surroundings and the energy that is used to heat the kettle itself.

THE ENERGY INVOLVED IN CHANGE OF STATE

When you supply energy to a substance you would expect its temperature to rise and this is generally true.

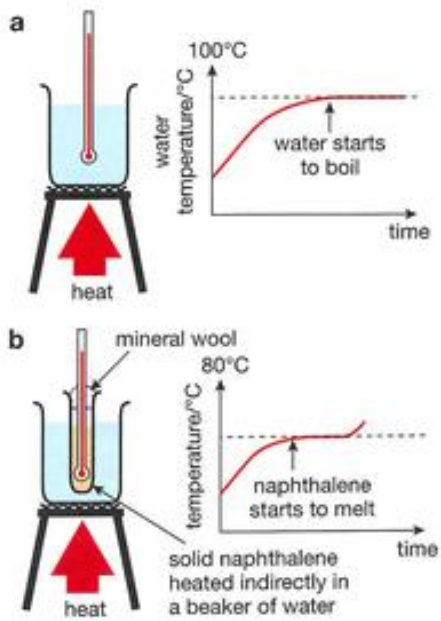
KEY POINT

Naphthalene is used because it has a lower melting point than many solids. It is flammable so must be heated indirectly in a water bath with care.

ACTIVITY 2

▼ PRACTICAL: INVESTIGATE TEMPERATURE DURING A CHANGE OF STATE

The following two experiments show situations where this is not the case.



▲ Figure 19.5 a Heating water; b Heating solid naphthalene

! Avoid skin contact with naphthalene and avoid inhaling its vapour. Stop heating when it has melted and do not remove the thermometer from the liquid or the solid.

Figure 19.5 shows two experiments involving a change of state. In Figure 19.5a water is being heated continuously. The water temperature rises until the water starts to boil and then remains constant until all the water has been turned from liquid to gas. The energy supplied is doing work separating the water molecules. In Figure 19.5b naphthalene (used to make moth balls), which has a melting point of 80 °C, is being heated continuously in a beaker of water. The temperature of the solid naphthalene rises until it starts to melt. Until all the naphthalene has melted, its temperature remains constant. The temperature will start to rise again when all the naphthalene is in the liquid state.

The thermal energy supplied is used to overcome the forces between the molecules of the liquid in the case of water and between the molecules of the solid in the case of naphthalene. If we allow the naphthalene to cool down, its temperature will drop until the naphthalene starts to turn back into a solid and will not start to fall again until all the liquid has turned back to solid. During any change of state, melting or freezing (solidifying), boiling or condensing (turning back from gas to liquid), the temperature remains constant.

END OF PHYSICS ONLY

THE GAS LAWS

We are now going to focus our attention on the properties of gases. We shall explain the different properties in terms of the movement of particles.

We have already said that gases are made up of molecules that are moving. We believe that the particles in gases are spread out and constantly moving in a random, haphazard way.

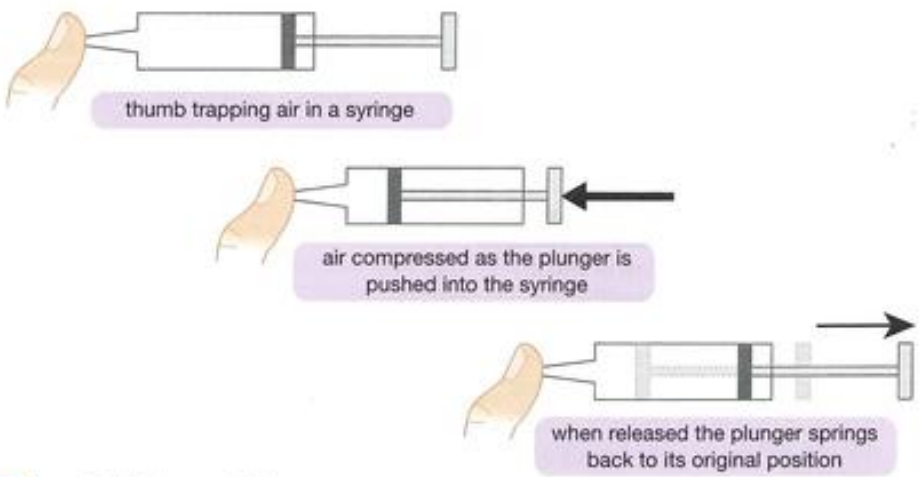
When the molecules hit the walls of a container they exert a force. The combined effect of the huge number of collisions results in the pressure that is exerted on the walls of the container.

BOYLE'S LAW



▲ Figure 19.6 Robert Boyle (1627–1691) was an Anglo-Irish chemist and philosopher. As well as discovering the law that bears his name, he worked with Robert Hooke at Oxford developing the air pump.

The scientist Robert Boyle (Figure 19.6) discovered something that you have probably noticed if you have ever used a bicycle pump: air is squashy! He noticed that you can squeeze air in a cylinder and that it springs back to its original volume when you release it. You can try this for yourself with a plastic syringe (Figure 19.7).



▲ Figure 19.7 Air is squashy!

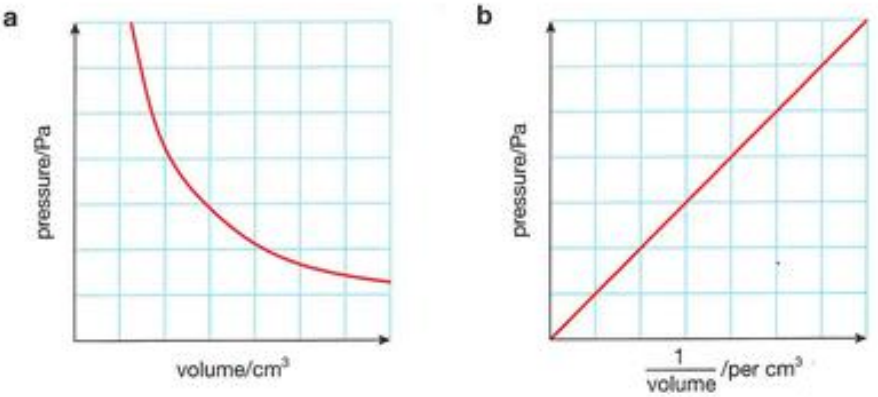
SOLIDS, LIQUIDS AND GASES



▲ Figure 19.8 A modern version of Boyle's experiment to see how the volume of a gas depends on the pressure exerted on it.

Boyle devised an experiment to see how the volume occupied by a gas depends on the pressure exerted on it. Pressure is the force acting per unit area. This is measured in N/m^2. One N/m^2 is called a pascal (Pa). A version of Boyle's experiment is shown in Figure 19.8.

Boyle took care to make sure that the trapped gas he was studying stayed at the same temperature. He increased the pressure on the gas and made a note of the new volume. His results looked like the graph shown in Figure 19.9a.



▲ Figure 19.9 a Graph to show how the pressure of a gas at constant temperature varies with the volume; b Graph of pressure against $\frac{1}{\text{volume}}$

Boyle noticed that when he doubled the pressure, the volume of the gas halved. If we plot pressure (p), against $\frac{1}{\text{volume}}$ $\left(\frac{1}{V}\right)$, as in Figure 19.9b, we can see from the straight line passing through the origin that p is proportional to $\frac{1}{V}$.

This discovery, called Boyle's law, is expressed in the equation:

$$p_1 V_1 = p_2 V_2$$

This means that if you take a fixed mass of gas that has a pressure p_1 and a volume V_1 and change either the pressure or the volume, the new pressure p_2 multiplied by the new volume V_2 will be the same as $p_1 \times V_1$, if the temperature of the gas remains the same.

!

Eye protection is needed in case any tubing leaks or detaches, which will cause gas under high pressure to be ejected.

EXAMPLE 4

Atmospheric pressure is 100 kPa (100 000 Pa). Some air in a sealed container has a volume of 2 m^3 at atmospheric pressure. What would be the pressure of the air if you reduced its volume to 0.2 m^3?

First write down what we know:

$$p_1 = 100 \text{ kPa}$$

$$V_1 = 2 \text{ m}^3$$

$$V_2 = 0.2 \text{ m}^3$$

$$p_1 V_1 = p_2 V_2$$

$$p_2 \times 0.2 \text{ m}^3 = 100 \text{ kPa} \times 2 \text{ m}^3$$

$$p_2 = \frac{100 \text{ kPa} \times 2 \text{ m}^3}{0.2 \text{ m}^3}$$

$$= 1000 \text{ kPa}$$

HINT

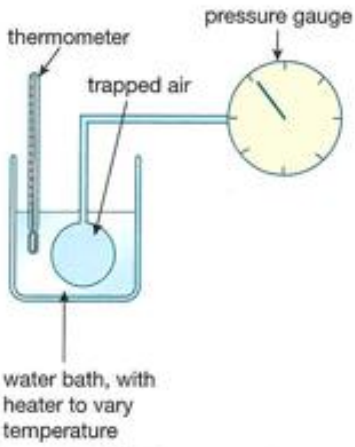
You can do all your working in kPa if you wish, rather than converting the pressure to pascals. You can also use cubic centimetres for volume if these units are given in the question. It does not matter which units you use, as long as you use the same ones throughout the question.

Gases can be compressed because the gas molecules are very spread out. When a gas is squashed into a smaller container it presses on the walls of the container with a greater pressure. This is explained in terms of particle theory as follows.

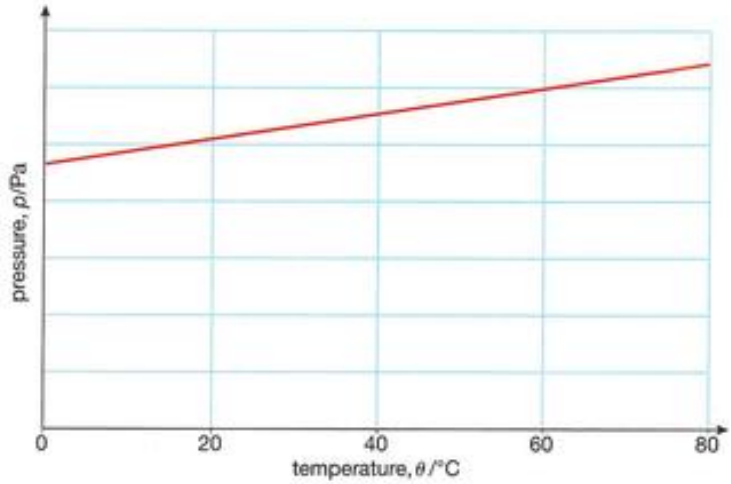
If the gas is kept at the same temperature, the average speed of the particles stays the same. (Remember that temperature is an indication of the kinetic energy of the particles.) If the same number of particles is squeezed into a smaller volume, they will hit the container walls more often. Each particle exerts a tiny force on the wall with which it collides. More collisions per second means a greater average force on the wall and, therefore, a greater pressure.

ABSOLUTE ZERO

Boyle took care to conduct his experiment at constant temperature. He was aware that temperature also had an effect on the pressure of a gas. Figure 19.10 shows an experiment to investigate how the pressure of a gas depends on its temperature.



▲ Figure 19.10 Simple apparatus to measure the pressure of a fixed volume of gas at a range of temperatures

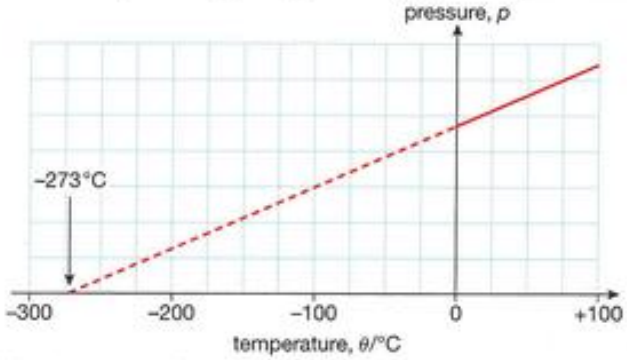


▲ Figure 19.11 Graph to show how the pressure of a fixed volume of gas varies with temperature

The gas is kept at constant volume, and its pressure is measured at a range of temperatures. The graph in Figure 19.11 shows typical results for this experiment.

The graph shows that the pressure of the gas increases as the temperature increases. We could also say that the gas pressure gets smaller as the gas is cooled.

What happens if we keep cooling the gas? The graph in Figure 19.12 shows this.



▲ Figure 19.12 At a temperature of −273 °C, the pressure of the gas would be zero. This temperature is known as 'absolute zero'.

As we cool the gas, the pressure keeps decreasing. The pressure of the gas cannot become less than zero. This suggests that there is a temperature below which it is not possible to cool the gas further. This temperature is called absolute zero. Experiments show that absolute zero is approximately –273 °C.

The Kelvin temperature scale starts from absolute zero. The Kelvin temperature of a gas is proportional to the average kinetic energy of its molecules.

To convert a temperature on the Celsius scale (in °C) to a Kelvin scale temperature (in K), add 273 to the Celsius scale temperature:

- temperature in K = temperature in °C + 273
- temperature in °C = temperature in K – 273

EXAMPLE 5

a At what temperature does water freeze, in Kelvin?

Water freezes at 0 °C. To convert 0 °C to Kelvin:

$T = (0\,°C + 273)\,K$

$= 273\,K$

b What is room temperature, in Kelvin?

Typical room temperature is 20 °C, so:

$T = (20\,°C + 273)\,K$

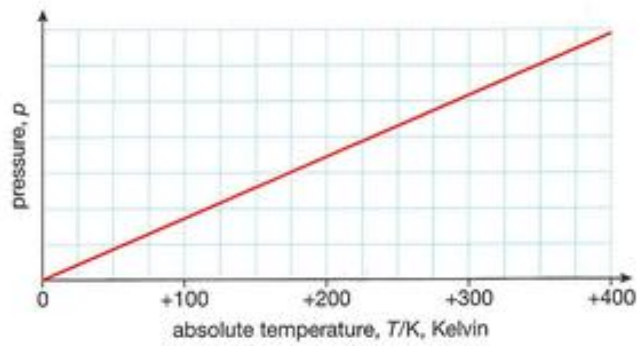
$= 293\,K$

c What temperature is 400 K on the Celsius scale?

$\theta = (400\,K - 273)\,°C$

$= 127\,°C$

If we redraw the graph of pressure against temperature using the absolute or Kelvin temperature scale, we get a graph that is a straight line passing through the origin, as shown in Figure 19.13. This shows that the pressure of the gas is proportional to its Kelvin temperature. For example, if you heat a gas from 200 K (–73 °C) to 400 K (127 °C), its pressure will double.



▲ Figure 19.13 Graph of pressure against absolute temperature for a fixed amount of gas at constant volume

KEY POINT

We often use the symbol θ for temperature in °C. For temperature in K we always use the symbol T. Remember to convert Celsius temperatures to Kelvin when using gas law equations in which temperature change occurs.

For a fixed mass of gas at a constant volume:

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

NB T must be measured in Kelvin.

EXAMPLE 6

You take an empty tin and put the lid on tightly. You heat it using a Bunsen burner until the temperature of the air inside is 50 °C. What is the pressure of the air inside the tin? The temperature of the room is 20 °C.

First write down what you know:

p_1 = 100 kPa (this is atmospheric pressure)

The starting temperature is 20 °C, but we must change this to Kelvin, so:

$T_1 = 20 + 273 \text{ K} = 293 \text{ K}$.

The end temperature, 50 °C, must also be converted so:

$T_2 = 50 + 273 \text{ K} = 323 \text{ K}$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

$$p_2 = p_1 \times \frac{T_2}{T_1}$$

$$= 100 \text{ kPa} \times \frac{323 \text{ K}}{293 \text{ K}}$$

$$= 110 \text{ kPa}$$

EXTENSION WORK

Cooling things further becomes more and more difficult as they get closer to 0 K. Scientists have managed to cool gases down to within a few thousandths of a degree above absolute zero. At this temperature, even gases like helium and hydrogen can be liquefied. The gas laws do not apply once a gas is either cooled or compressed (forced into a small volume) to the point that they become liquid.

The relationship can be explained as follows.

The number of gas particles and the space, or volume, they occupy remain constant. When we heat the gas the particles continue to move randomly, but with a higher average speed. This means that their collisions with the walls of the container are harder and happen more often. This results in the average pressure exerted by the particles increasing.

When we cool a gas the kinetic energy of its particles decreases. The lower the temperature of a gas the less kinetic energy its particles have – they move more slowly. At absolute zero the particles have no thermal or movement energy, so they cannot exert a pressure.

CHAPTER QUESTIONS

More questions on solids, liquids and gases can be found at the end of Unit 5 on page 193.

PHYSICS ONLY

SKILLS INTERPRETATION 5

SKILLS CRITICAL THINKING 6

1 a Draw up a table to summarise the way that particles are arranged in solids, liquids and gases.

b How do these different arrangements of particles explain the physical properties of solids, liquids and gases?

2 Copy and complete the following paragraph about the particle theory of matter.

Matter is made up of _____ that are in continuous _____. When we supply heat energy to matter the _____ move _____. Gases exert a pressure on their containers because the _____ are continually _____ with the walls. The pressure will_____ when the gas is heated in a container of fixed volume because the _____ are moving _____.

END OF PHYSICS ONLY

SKILLS CRITICAL THINKING (6)

3 Explain how ideas about particles can account for the absolute zero of temperature.

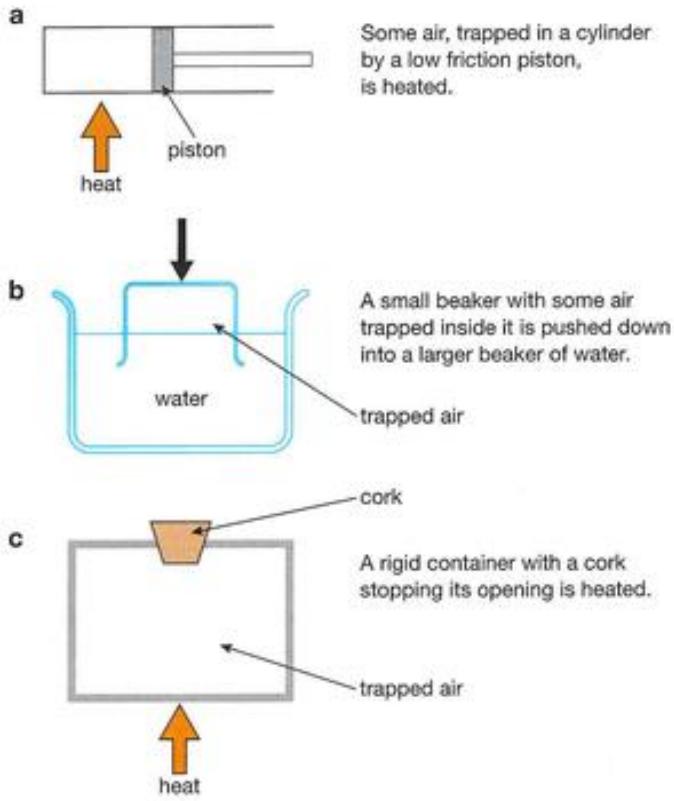
SKILLS PROBLEM SOLVING (5)

4 a Convert the following Celsius temperatures to Kelvin temperatures.
- i 0 °C
- ii 100 °C
- iii 20 °C

b Convert the following Kelvin temperatures to Celsius temperatures.
- i 250 K
- ii 269 K
- iii 305 K

SKILLS ANALYSIS (6)

5 State what happens in the situations shown in the diagram. Explain your answers using ideas about particles.



PHYSICS ONLY

SKILLS PROBLEM SOLVING (8)

6 a A fixed amount of gas occupies a volume of 500 m^3 when under a pressure of 100 kPa. Calculate the volume that this amount of gas will occupy when its pressure is increased to 125 kPa. You should show each stage of your calculation.

SKILLS REASONING (6)

b What do you need to assume about the temperature of the gas? Explain your answer.

SKILLS PROBLEM SOLVING

7 a You put the lid on a jar in a kitchen where the air temperature is 20 °C. The jar gets left out in the winter, and the air temperature drops to –5 °C. Explain what happens to the air pressure in the cold jar.

SKILLS REASONING (7)

b Explain why it might be more difficult to remove the lid of the jar when it is cold.

END OF PHYSICS ONLY